2. Estimation of Ferrous Ammonium Sulphate (FAS)

Aim :

To estimate the amount of ferrous ammonium sulphate (FAS) dissolved in 1500 ml of the given unknown solution volumetrically. For this you are given with a standard solution of ferrous sulphate ($FeSO_4$)of normality 0.1024 N and potassium permanganate solution as link solution.

Principle:

During these titrations, Fe^{3+} ions (from ferrous salts) are oxidised to Fe^{3+} ions and $5\,Fe^{2+} \longrightarrow 5\,Fe^{3+} + 5e^-$ ion (from $5\,Fe^{2+} \longrightarrow 5\,Fe^{3+} + 5e^-$) is reduced to $5\,Fe^{2+} \longrightarrow 5\,Fe^{3+} + 5e^-$ ion.

Oxidation : $5\,Fe^{2+} \longrightarrow 5\,Fe^{3+} + 5e^-$

Reduction : $5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$

Overall reaction : $5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$

Short procedure:

s.no	Content	Titration-I	Titration-II
1	Burette solution	$KMnO_4$	$KMnO_4$
2	Pipette solution	20 ml of standard $FeSO_4$	20 ml of unknown FAS
3	Acid to be added	20ml of 2N H_2SO_4 (approx)	20ml of 2N H_2SO_4 (approx)
4	Temperature	Lab temperature	Lab temperature
5	Indicator	Self-indicator ($KMnO_4$)	Self-indicator ($KMnO_4$)
6	End point	Appearance of permanent pale pink colour	Appearance of permanent pale pink colour
7	Equivalent weight of FAS = 392		

Procedure :

Titration–I

(Link $FeSO_4$)Vs (Standard $FeSO_4$)

Burette is washed with water, rinsed with $FeSO_4$ solution and filled with same $FeSO_4$ solution up to the zero mark. Exactly 20 ml of standard $FeSO_4$ solution is pipetted out into the clean, washed conical flask. To this $KMnO_4$ solution, approximately 20ml of 2N sulphuric acid is added. This mixture is titrated against $KMnO_4$ Link solution from the burette. $KMnO_4$ is added drop wise till the appearance of permanent pale pink colour. Burette reading are noted, the same procedure is repeated to get concordant values.

organic analysis.indd 268 3/29/2019 6:48:25 PM